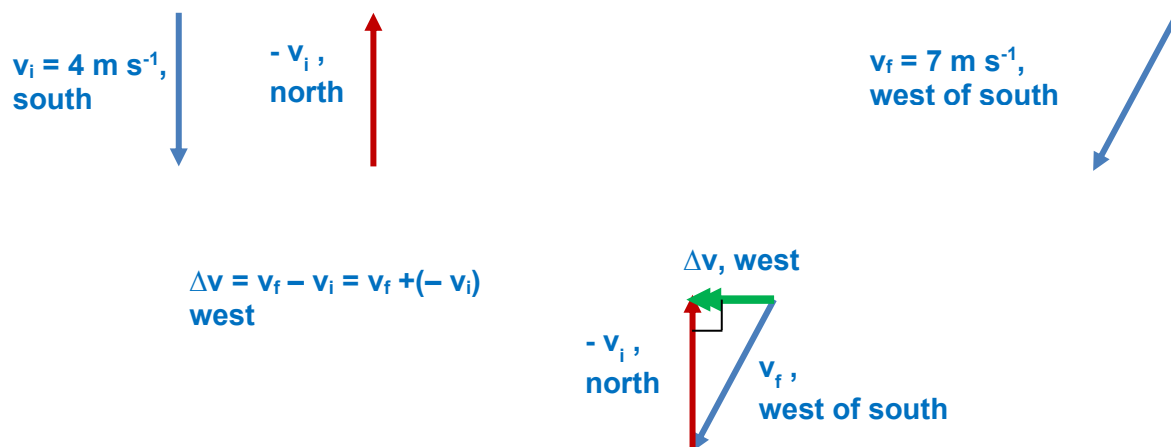


3	A boat moves at velocity of 4 m s^{-1} due south in still water across a lake. It then experiences a current flowing towards the west. The velocity of the boat changes to 7 m s^{-1}. What is the change in the velocity of the boat due to the current?	
	A	3 m s^{-1} due west
	B	3 m s^{-1} due west of south
	C	6 m s^{-1} due west
	D	6 m s^{-1} due west of south

Answer: C

Boat's final velocity must be due west of south.

Direction of change in velocity must be that of the current's direction, i.e. due west.



Magnitude of change in velocity, $|\Delta v| = \sqrt{|v_f|^2 - |v_i|^2} = \sqrt{7^2 - 4^2} = 6 \text{ m s}^{-1}$

5

4 Two cars are initially side by side and at rest. They then accelerate in the same direction along